RE-CHARACTERIZATION OF UNIVERSAL CLOSEDNESS VIA THE REASONABLE STRUCTURE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- First characterization. proposition-characterize-universally-closed (carded L26735). The valuative formulation.
- Degree stratification. lemma-quasi-compact-reasonable-stratification (carded L26710). The stratification of the fibres.

Source reference. Original lemma 03M5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3910–3910. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 03KJ: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3789–3789. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KA: focused-seeds/geometry/S9-decent-re-characterize-universally-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-compact and decent. (For example if f is representable, or quasi-separated, see Lemma 03M5.) Then f is universally closed if and only if the existence part of the valuative criterion holds.*

Proof. In Morphisms of Spaces, Lemma 03KA we proved that any quasi-compact morphism which satisfies the existence part of the valuative criterion is universally closed. To prove the other, assume that f is universally closed. In the proof of Proposition 03KJ we have seen that it suffices to show, for any valuation ring A, and any morphism $\mathrm{Spec}(A) \to Y$, that the base change $f_A : X_A \to \mathrm{Spec}(A)$ satisfies the existence part of the valuative criterion. By definition the algebraic space X_A has property (γ) and hence Proposition 03KJ applies to the morphism f_A and we win. □

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